Quiz 3: Practice Solutions

Lectures 8–11

Try the practice handout in 15 minutes before reading these solutions. Each part earns 2 practice points for a correct final answer and 0 otherwise. The quiz covers the same lectures and skills in different problems.

1(a)	1(b)	2(a)	2(b)	2(c)
5	4	C	B	B

1. One update, two targets — 4 points

(a) Q-learning: $Q(\mathrm{X}, a) = 5$

Q-learning uses the largest next-state value, $\max\{4, 8\} = 8$:

$$Q(\mathrm{X}, a) \leftarrow 2 + \tfrac{1}{2}\big(4 + \tfrac{1}{2}\max\{4, 8\} - 2\big) = \boxed{5}.$$

(b) SARSA: $Q(\mathrm{X}, a) = 4$

Start again from $Q(\mathrm{X}, a) = 2$. SARSA uses the chosen next action a, so it reads $Q(\mathrm{Y}, a) = 4$:

$$Q(\mathrm{X}, a) \leftarrow 2 + \tfrac{1}{2}\big(4 + \tfrac{1}{2} \cdot 4 - 2\big) = \boxed{4}.$$

2. Estimates, targets, and policies — 6 points

(a) C: Q-learning only

$$\begin{aligned} \text{SARSA:} \quad & Q^{\pi_b}(s, a) = r(s, a) + \gamma \textstyle\sum_{s'} P(s' \mid s, a) \sum_{a'} \pi_b(a' \mid s') Q^{\pi_b}(s', a'), \\ \text{Q-learning:} \quad & Q^{\star}(s, a) = r(s, a) + \gamma \textstyle\sum_{s'} P(s' \mid s, a) \max_{a'} Q^{\star}(s', a'). \end{aligned}$$

SARSA's is the Bellman expectation equation for π_b, not the optimality equation. Hence $Q^\star$ need not satisfy it; the guarantee holds only for Q-learning.

(b) B: $V^\pi(s) + \gamma P^\pi(s_0 \mid s)\, c$

The mean of the target is the backup of the current table; the table's error enters only through the next state:

$$\mathbb{E}\big[R_{t+1} + \gamma V(S_{t+1}) \mid S_t = s\big] = r^\pi(s) + \gamma \sum_{s'} P^\pi(s' \mid s)\, V(s') = V^\pi(s) + \gamma P^\pi(s_0 \mid s)\, c,$$

also when $s = s_0$. A omits the transition probability; C omits a bias that need not vanish (it does if $P^\pi(s_0 \mid s) = 0$); D drops the discount.

(c) B: I only

I. Since every state is visited and $\alpha > 0$, unchanged updates give the Bellman equation:

$$Q(s, \pi_g(s)) = r(s, \pi_g(s)) + \gamma Q(s', \pi_g(s')).$$

Here s' is the next state. Expanding along π_g from $s_0 = s$ ($\gamma < 1$):

$$Q(s, \pi_g(s)) = \sum_{t=0}^{\infty} \gamma^t r(s_t, \pi_g(s_t)) = Q^{\pi_g}(s, \pi_g(s)) = V^{\pi_g}(s).$$

II. Visiting all states need not visit all actions. One state, two self-looping actions, $\gamma = \frac{1}{2}$:

$$r(s, a) = 0, \quad r(s, b) = 1, \qquad Q(s, a) = 0, \quad Q(s, b) = -1.$$

Only a is chosen, and its target is $0 + \frac{1}{2}\max\{0, -1\} = 0$. Thus Q never changes and b is never tried, but $V^{\pi_g}(s) = 0 < V^\star(s) = 2$.